

LOSO (glove_ml_explorer) Experiment SB_accel

Static · accelerometers only (both hands), no Nothing

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Experiment configuration

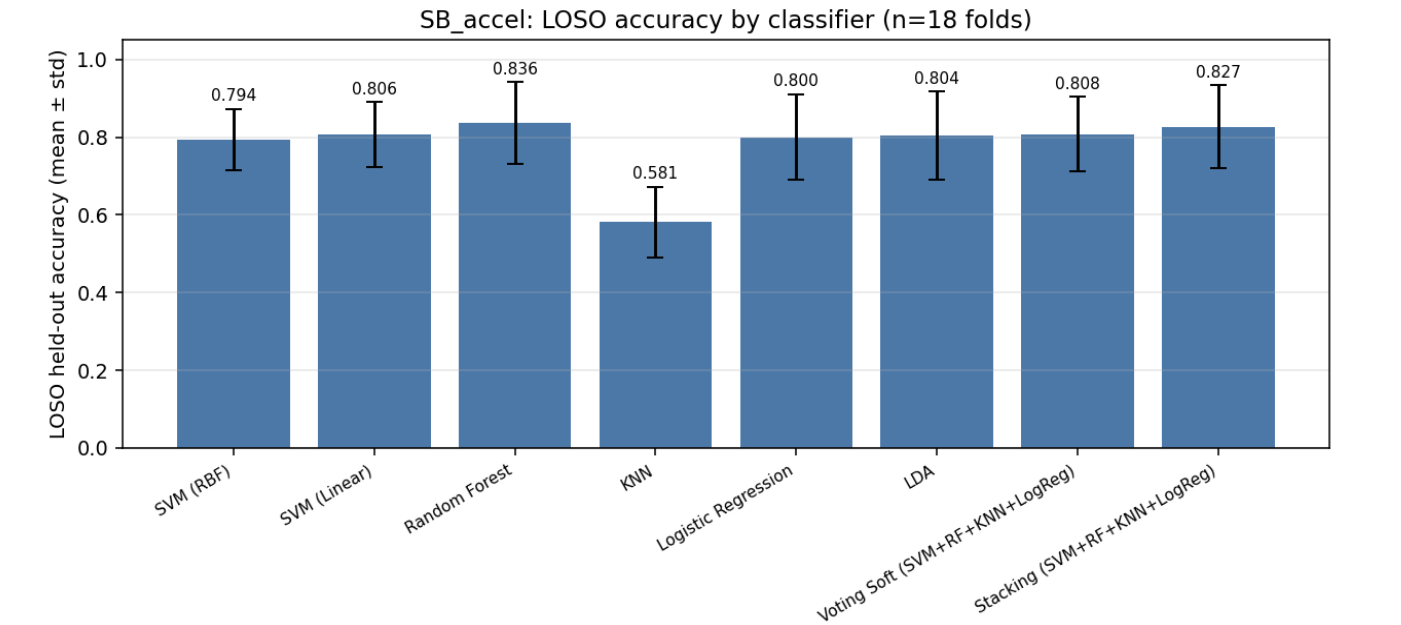
Class set	6 classes excluding Double_Nothing
Classes	Double_Closed_Fist, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman
Sensor modalities	accel
Hands	left, right
Segments	thumb, index, middle, ring, pinky, wrist
Sensor channels	66
Sequence length	90 steps (resampled)
Filter	off
Feature mode	stats+fft
Normalisation	minmax
Augmentation	off
Inner CV folds	5
LOSO subjects	18: 10_Josh, 11_Mansh, 12_Marcus, 13_Nat, 14_StephenV2, 16_Bella, 17_Raquel, 1_Alan, 2_Alex, 3_Anghad, 4_Daniel, 5_Georgia, 6_Harry, 7_Henry, 8_Jestin, 9_Joselyn, Stephen, Val

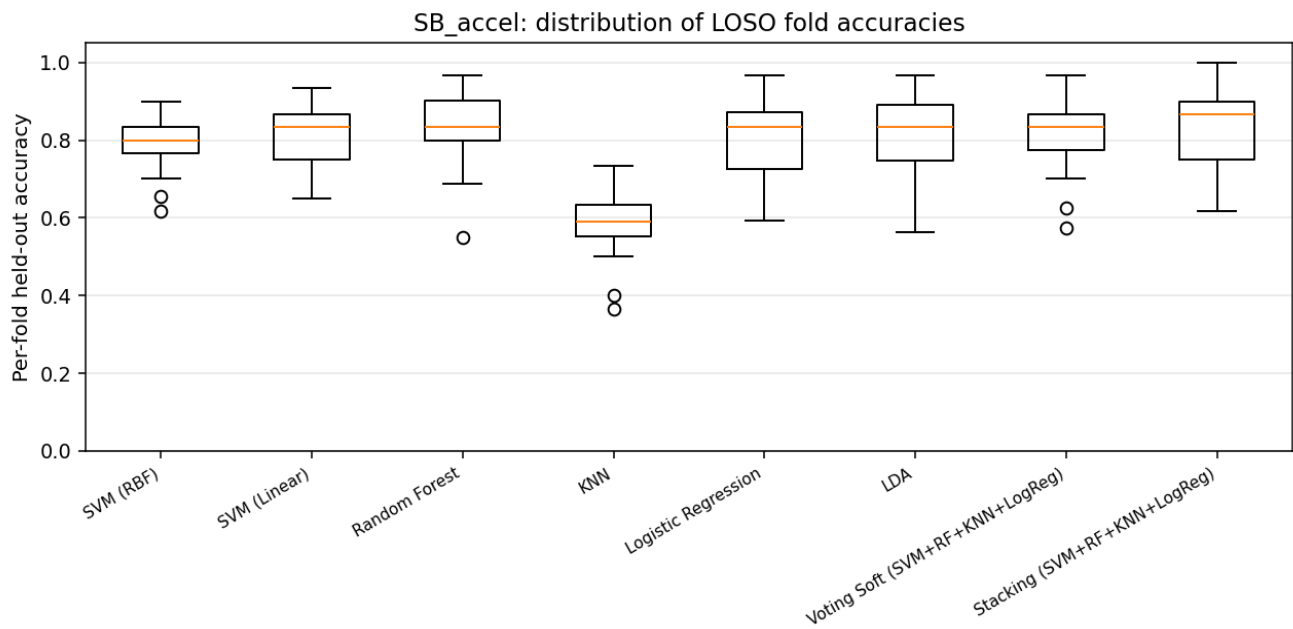
LOSO summary across all subjects

Classifier	Params*	LOSO mean	LOSO std	Min fold	Max fold	Inner CV mean
SVM (RBF)	—	0.7937	0.0793	0.6167	0.9000	0.8150
SVM (Linear)	53,475	0.8064	0.0834	0.6500	0.9333	0.8426
Random Forest	26,532	0.8360	0.1058	0.5500	0.9667	0.8878
KNN	—	0.5811	0.0914	0.3667	0.7333	0.5771
Logistic Regression	21,390	0.8001	0.1109	0.5938	0.9667	0.8240
LDA	21,390	0.8039	0.1136	0.5625	0.9667	0.8007
Voting Soft (SVM+RF+KNN+LogReg)	—	0.8078	0.0960	0.5750	0.9667	0.8411
Stacking (SVM+RF+KNN+LogReg)	—	0.8267	0.1067	0.6167	1.0000	0.8650

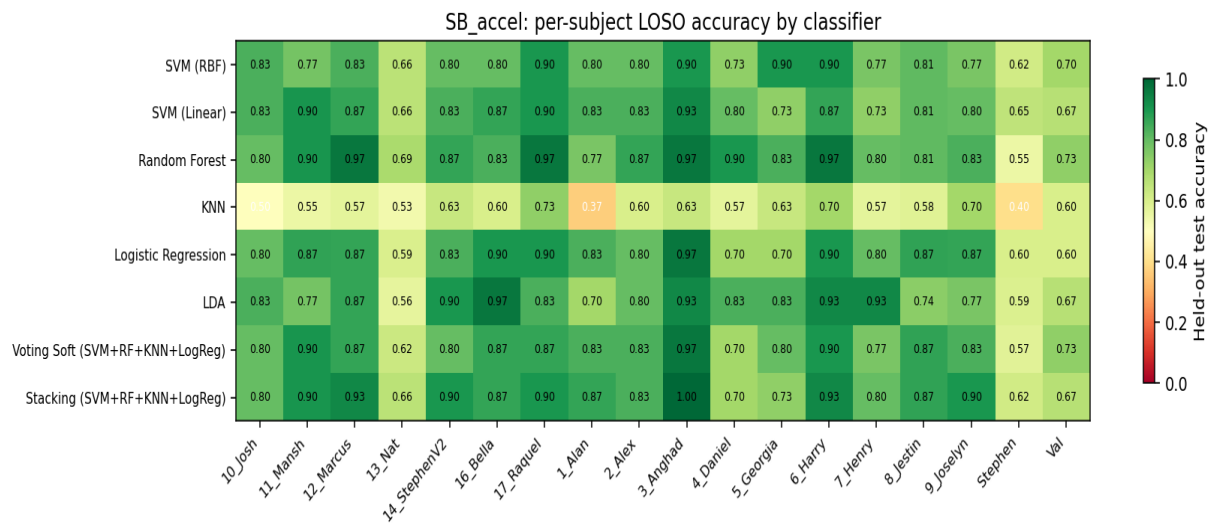
*Params: linear models use coef_+intercept size; tree ensembles use total node count across trees. SVM-RBF / KNN / Voting / Stacking are non-parametric or composite, so they show as —.

Best classifier by LOSO mean: **Random Forest**





Per-subject LOSO accuracy heatmap



Per-fold detail (held-out subject = each row)

Held-out subj	Train trials	SVM (RBF)	SVM (Linear)	Random Forest	KNN	Logistic Regression	LDA	Voting Soft (SVM+RF+KNN+LogReg)	Stacking (SVM+RF+KNN+LogReg)
10_Josh	604	0.833	0.833	0.800	0.500	0.800	0.833	0.800	0.800
11_Mansh	603	0.774	0.903	0.903	0.548	0.871	0.774	0.903	0.903
12_Marcus	604	0.833	0.867	0.967	0.567	0.867	0.867	0.867	0.933
13_Nat	602	0.656	0.656	0.688	0.531	0.594	0.562	0.625	0.656
14_StephenV2	604	0.800	0.833	0.867	0.633	0.833	0.900	0.800	0.900
16_Bella	604	0.800	0.867	0.833	0.600	0.900	0.967	0.867	0.867
17_Raquel	604	0.900	0.900	0.967	0.733	0.900	0.833	0.867	0.900
1_Alan	604	0.800	0.833	0.767	0.367	0.833	0.700	0.833	0.867
2_Alex	604	0.800	0.833	0.867	0.600	0.800	0.800	0.833	0.833
3_Anghad	604	0.900	0.933	0.967	0.633	0.967	0.933	0.967	1.000
4_Daniel	604	0.733	0.800	0.900	0.567	0.700	0.833	0.700	0.700
5_Georgia	604	0.900	0.733	0.833	0.633	0.700	0.833	0.800	0.733
6_Harry	604	0.900	0.867	0.967	0.700	0.900	0.933	0.900	0.933
7_Henry	604	0.767	0.733	0.800	0.567	0.800	0.933	0.767	0.800
8_Jestin	603	0.806	0.806	0.806	0.581	0.871	0.742	0.871	0.871
9_Joselyn	604	0.767	0.800	0.833	0.700	0.867	0.767	0.833	0.900
Stephen	514	0.617	0.650	0.550	0.400	0.600	0.592	0.575	0.617
Val	604	0.700	0.667	0.733	0.600	0.600	0.667	0.733	0.667

Classifier: SVM (RBF)

Params (best-effort): —

LOSO mean accuracy: 0.7937 ± 0.0793 (min 0.6167, max 0.9000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	1/5	5/5	5/5	4/5	5/5	5/5	0.833
11_Mansh	2/6	3/5	5/5	5/5	4/5	5/5	0.774
12_Marcus	1/5	4/5	5/5	5/5	5/5	5/5	0.833
13_Nat	4/6	1/5	5/5	0/5	6/6	5/5	0.656
14_StephenV2	2/5	5/5	4/5	3/5	5/5	5/5	0.800
16_Bella	5/5	5/5	3/5	1/5	5/5	5/5	0.800
17_Raquel	5/5	4/5	5/5	3/5	5/5	5/5	0.900
1_Alan	2/5	5/5	4/5	4/5	4/5	5/5	0.800
2_Alex	3/5	5/5	5/5	1/5	5/5	5/5	0.800
3_Anghad	3/5	5/5	5/5	4/5	5/5	5/5	0.900
4_Daniel	4/5	5/5	2/5	1/5	5/5	5/5	0.733
5_Georgia	4/5	5/5	4/5	4/5	5/5	5/5	0.900
6_Harry	5/5	3/5	5/5	4/5	5/5	5/5	0.900
7_Henry	3/5	3/5	5/5	2/5	5/5	5/5	0.767
8_Jestin	4/6	4/5	4/5	3/5	5/5	5/5	0.806
9_Joselyn	4/5	5/5	4/5	1/5	5/5	4/5	0.767
Stephen	3/20	11/20	17/20	13/20	12/20	18/20	0.617
Val	2/5	3/5	4/5	3/5	5/5	4/5	0.700
MEAN	0.590	0.808	0.869	0.569	0.956	0.972	0.794

Classifier: SVM (Linear)

Params (best-effort): 53,475

LOSO mean accuracy: 0.8064 ± 0.0834 (min 0.6500, max 0.9333)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	1/5	5/5	5/5	4/5	5/5	5/5	0.833
11_Mansh	4/6	4/5	5/5	5/5	5/5	5/5	0.903
12_Marcus	2/5	4/5	5/5	5/5	5/5	5/5	0.867
13_Nat	4/6	1/5	5/5	0/5	6/6	5/5	0.656
14_StephenV2	2/5	5/5	4/5	4/5	5/5	5/5	0.833
16_Bella	5/5	5/5	5/5	1/5	5/5	5/5	0.867
17_Raquel	5/5	4/5	5/5	3/5	5/5	5/5	0.900
1_Alan	2/5	5/5	5/5	4/5	4/5	5/5	0.833
2_Alex	4/5	5/5	5/5	1/5	5/5	5/5	0.833
3_Anghad	5/5	5/5	5/5	3/5	5/5	5/5	0.933
4_Daniel	4/5	5/5	3/5	2/5	5/5	5/5	0.800
5_Georgia	0/5	5/5	5/5	2/5	5/5	5/5	0.733
6_Harry	5/5	4/5	5/5	2/5	5/5	5/5	0.867
7_Henry	3/5	2/5	5/5	2/5	5/5	5/5	0.733
8_Jestin	5/6	5/5	4/5	2/5	4/5	5/5	0.806
9_Joselyn	4/5	5/5	3/5	3/5	5/5	4/5	0.800
Stephen	4/20	13/20	17/20	16/20	11/20	17/20	0.650
Val	2/5	2/5	5/5	2/5	5/5	4/5	0.667
MEAN	0.620	0.825	0.925	0.544	0.953	0.969	0.806

Classifier: Random Forest

Params (best-effort): 26,532

LOSO mean accuracy: 0.8360 ± 0.1058 (min 0.5500, max 0.9667)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	1/5	5/5	5/5	3/5	5/5	5/5	0.800
11_Mansh	4/6	4/5	5/5	5/5	5/5	5/5	0.903
12_Marcus	4/5	5/5	5/5	5/5	5/5	5/5	0.967
13_Nat	5/6	1/5	5/5	0/5	6/6	5/5	0.688
14_StephenV2	5/5	4/5	5/5	2/5	5/5	5/5	0.867
16_Bella	4/5	5/5	5/5	1/5	5/5	5/5	0.833
17_Raquel	5/5	5/5	5/5	4/5	5/5	5/5	0.967
1_Alan	5/5	3/5	5/5	0/5	5/5	5/5	0.767
2_Alex	5/5	5/5	5/5	1/5	5/5	5/5	0.867
3_Anghad	5/5	5/5	5/5	4/5	5/5	5/5	0.967
4_Daniel	5/5	5/5	5/5	2/5	5/5	5/5	0.900
5_Georgia	2/5	5/5	5/5	3/5	5/5	5/5	0.833
6_Harry	5/5	5/5	5/5	4/5	5/5	5/5	0.967
7_Henry	3/5	2/5	5/5	4/5	5/5	5/5	0.800
8_Jestin	5/6	5/5	5/5	0/5	5/5	5/5	0.806
9_Joselyn	5/5	5/5	5/5	0/5	5/5	5/5	0.833
Stephen	4/20	14/20	11/20	4/20	15/20	18/20	0.550
Val	1/5	4/5	5/5	3/5	5/5	4/5	0.733
MEAN	0.752	0.850	0.975	0.467	0.986	0.983	0.836

Classifier: KNN

Params (best-effort): —

LOSO mean accuracy: 0.5811 ± 0.0914 (min 0.3667, max 0.7333)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	0/5	0/5	5/5	1/5	5/5	4/5	0.500
11_Mansh	1/6	2/5	3/5	3/5	4/5	4/5	0.548
12_Marcus	0/5	0/5	5/5	2/5	5/5	5/5	0.567
13_Nat	2/6	1/5	4/5	0/5	6/6	4/5	0.531
14_StephenV2	2/5	4/5	2/5	1/5	5/5	5/5	0.633
16_Bella	3/5	2/5	3/5	0/5	5/5	5/5	0.600
17_Raquel	4/5	3/5	5/5	2/5	4/5	4/5	0.733
1_Alan	3/5	0/5	2/5	1/5	3/5	2/5	0.367
2_Alex	3/5	1/5	4/5	3/5	3/5	4/5	0.600
3_Anghad	3/5	2/5	3/5	3/5	5/5	3/5	0.633
4_Daniel	3/5	1/5	3/5	1/5	4/5	5/5	0.567
5_Georgia	3/5	4/5	2/5	1/5	5/5	4/5	0.633
6_Harry	4/5	1/5	3/5	3/5	5/5	5/5	0.700
7_Henry	3/5	4/5	2/5	0/5	4/5	4/5	0.567
8_Jestin	3/6	3/5	4/5	1/5	5/5	2/5	0.581
9_Joselyn	3/5	3/5	4/5	1/5	5/5	5/5	0.700
Stephen	5/20	4/20	10/20	3/20	8/20	18/20	0.400
Val	2/5	2/5	5/5	2/5	4/5	3/5	0.600
MEAN	0.469	0.378	0.683	0.286	0.867	0.806	0.581

Classifier: Logistic Regression

Params (best-effort): 21,390

LOSO mean accuracy: 0.8001 ± 0.1109 (min 0.5938, max 0.9667)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	0/5	5/5	5/5	4/5	5/5	5/5	0.800
11_Mansh	4/6	4/5	5/5	4/5	5/5	5/5	0.871
12_Marcus	2/5	4/5	5/5	5/5	5/5	5/5	0.867
13_Nat	2/6	1/5	5/5	0/5	6/6	5/5	0.594
14_StephenV2	2/5	5/5	4/5	4/5	5/5	5/5	0.833
16_Bella	5/5	5/5	5/5	2/5	5/5	5/5	0.900
17_Raquel	5/5	4/5	5/5	3/5	5/5	5/5	0.900
1_Alan	1/5	5/5	5/5	4/5	5/5	5/5	0.833
2_Alex	4/5	5/5	5/5	0/5	5/5	5/5	0.800
3_Anghad	5/5	5/5	4/5	5/5	5/5	5/5	0.967
4_Daniel	3/5	5/5	3/5	0/5	5/5	5/5	0.700
5_Georgia	1/5	4/5	4/5	2/5	5/5	5/5	0.700
6_Harry	5/5	4/5	5/5	3/5	5/5	5/5	0.900
7_Henry	3/5	3/5	5/5	3/5	5/5	5/5	0.800
8_Jestin	5/6	5/5	5/5	2/5	5/5	5/5	0.871
9_Joselyn	5/5	5/5	3/5	3/5	5/5	5/5	0.867
Stephen	4/20	12/20	17/20	9/20	12/20	18/20	0.600
Val	2/5	2/5	5/5	2/5	4/5	3/5	0.600
MEAN	0.591	0.822	0.914	0.536	0.967	0.972	0.800

Classifier: LDA

Params (best-effort): 21,390

LOSO mean accuracy: 0.8039 ± 0.1136 (min 0.5625, max 0.9667)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	2/5	5/5	4/5	4/5	5/5	5/5	0.833
11_Mansh	2/6	3/5	5/5	4/5	5/5	5/5	0.774
12_Marcus	2/5	4/5	5/5	5/5	5/5	5/5	0.867
13_Nat	2/6	0/5	5/5	0/5	6/6	5/5	0.562
14_StephenV2	4/5	5/5	4/5	4/5	5/5	5/5	0.900
16_Bella	5/5	5/5	5/5	4/5	5/5	5/5	0.967
17_Raquel	5/5	4/5	4/5	2/5	5/5	5/5	0.833
1_Alan	1/5	4/5	5/5	3/5	5/5	3/5	0.700
2_Alex	4/5	4/5	5/5	1/5	5/5	5/5	0.800
3_Anghad	4/5	5/5	5/5	4/5	5/5	5/5	0.933
4_Daniel	5/5	5/5	3/5	2/5	5/5	5/5	0.833
5_Georgia	2/5	5/5	4/5	4/5	5/5	5/5	0.833
6_Harry	5/5	5/5	4/5	4/5	5/5	5/5	0.933
7_Henry	4/5	4/5	5/5	5/5	5/5	5/5	0.933
8_Jestin	2/6	5/5	4/5	5/5	2/5	5/5	0.742
9_Joselyn	4/5	5/5	3/5	2/5	5/5	4/5	0.767
Stephen	5/20	15/20	14/20	10/20	10/20	17/20	0.592
Val	2/5	4/5	2/5	4/5	5/5	3/5	0.667
MEAN	0.614	0.842	0.839	0.661	0.939	0.936	0.804

Classifier: Voting Soft (SVM+RF+KNN+LogReg)

Params (best-effort): —

LOSO mean accuracy: 0.8078 ± 0.0960 (min 0.5750, max 0.9667)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	0/5	5/5	5/5	4/5	5/5	5/5	0.800
11_Mansh	4/6	4/5	5/5	5/5	5/5	5/5	0.903
12_Marcus	2/5	4/5	5/5	5/5	5/5	5/5	0.867
13_Nat	3/6	1/5	5/5	0/5	6/6	5/5	0.625
14_StephenV2	2/5	5/5	4/5	3/5	5/5	5/5	0.800
16_Bella	5/5	5/5	5/5	1/5	5/5	5/5	0.867
17_Raquel	5/5	4/5	5/5	2/5	5/5	5/5	0.867
1_Alan	2/5	5/5	5/5	3/5	5/5	5/5	0.833
2_Alex	4/5	5/5	5/5	1/5	5/5	5/5	0.833
3_Anghad	5/5	5/5	5/5	4/5	5/5	5/5	0.967
4_Daniel	3/5	5/5	3/5	0/5	5/5	5/5	0.700
5_Georgia	2/5	5/5	4/5	3/5	5/5	5/5	0.800
6_Harry	5/5	4/5	5/5	3/5	5/5	5/5	0.900
7_Henry	3/5	3/5	5/5	2/5	5/5	5/5	0.767
8_Jestin	5/6	5/5	5/5	2/5	5/5	5/5	0.871
9_Joselyn	5/5	5/5	4/5	1/5	5/5	5/5	0.833
Stephen	3/20	13/20	17/20	7/20	10/20	19/20	0.575
Val	3/5	3/5	5/5	2/5	5/5	4/5	0.733
MEAN	0.631	0.847	0.936	0.475	0.972	0.986	0.808

Classifier: Stacking (SVM+RF+KNN+LogReg)

Params (best-effort): —

LOSO mean accuracy: 0.8267 ± 0.1067 (min 0.6167, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	0/5	5/5	5/5	4/5	5/5	5/5	0.800
11_Mansh	4/6	4/5	5/5	5/5	5/5	5/5	0.903
12_Marcus	3/5	5/5	5/5	5/5	5/5	5/5	0.933
13_Nat	4/6	1/5	5/5	0/5	6/6	5/5	0.656
14_StephenV2	3/5	5/5	5/5	4/5	5/5	5/5	0.900
16_Bella	5/5	5/5	5/5	1/5	5/5	5/5	0.867
17_Raquel	5/5	4/5	5/5	3/5	5/5	5/5	0.900
1_Alan	3/5	5/5	5/5	3/5	5/5	5/5	0.867
2_Alex	4/5	5/5	5/5	1/5	5/5	5/5	0.833
3_Anghad	5/5	5/5	5/5	5/5	5/5	5/5	1.000
4_Daniel	3/5	5/5	3/5	0/5	5/5	5/5	0.700
5_Georgia	1/5	5/5	4/5	2/5	5/5	5/5	0.733
6_Harry	5/5	4/5	5/5	4/5	5/5	5/5	0.933
7_Henry	3/5	3/5	5/5	3/5	5/5	5/5	0.800
8_Jestin	5/6	5/5	5/5	2/5	5/5	5/5	0.871
9_Joselyn	5/5	5/5	5/5	2/5	5/5	5/5	0.900
Stephen	3/20	12/20	16/20	15/20	9/20	19/20	0.617
Val	2/5	4/5	5/5	2/5	4/5	3/5	0.667
MEAN	0.651	0.867	0.956	0.553	0.958	0.975	0.827